

INTERMEDIATE MACROECONOMICS: ECN 2215



Money Markets

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THE DEMAND FOR MONEY



FINANCIAL WEALTH

DETERMINANTS OF ALLOCATION:

- LEVEL OF TRANSACTIONS
- INTEREST RATE ON BONDS

The demand for money = the amount people want to hold in money (M^d)

M^d in economy = $\sum$ individual demands for money in the economy

ALLOCATION

MONEY

INTEREST = 0

USED FOR TRANSACTIONS

TWO TYPES:

CURRENCY

CHECKABLE DEPOSITS

BONDS

INTEREST > 0

NOT USED FOR TRANSACTIONS

Supply and Demand for Money

- Demand for money depends on:

- Transactions motive
- Precautionary motive
- Speculative motive

1. Transactions Demand for Money

- ✦ Aimed at meeting current purchase of goods and services
- ✦ It is a function of the level of income and this relationship is positive
- ✦ Higher income – the greater the transactions demand for money
- ✦ It also depends on interest rate which is the opportunity cost of holding money for transactions purposes
- ✦ This demand for money and interest rate are negatively related. The higher the rate of interest the less will be the transactions demand for money

Supply and Demand for Money



2. Precautionary Demand for money

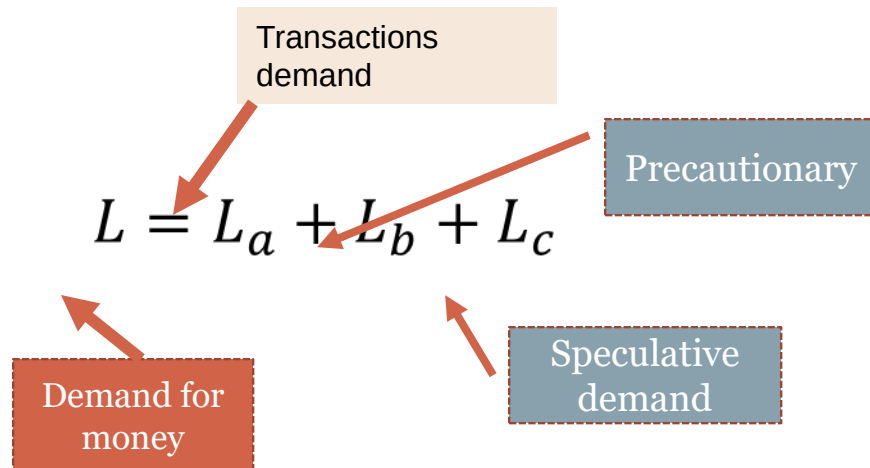
- This is the demand for money to meet unforeseen future needs for goods and services
- It's a function of income and the relationship is positive
- The rate of interest is the OC for demand of money for precautionary purposes

3. Speculative demand for money

- This is for speculative purposes to purchase stocks and bonds
- It's a function of the rate of interest and the relationship is negative

Supply and Demand for Money

Money demand Function



$$L = L_a(Y, r) + L_b(y, r) + L_c$$

Md depends upon:

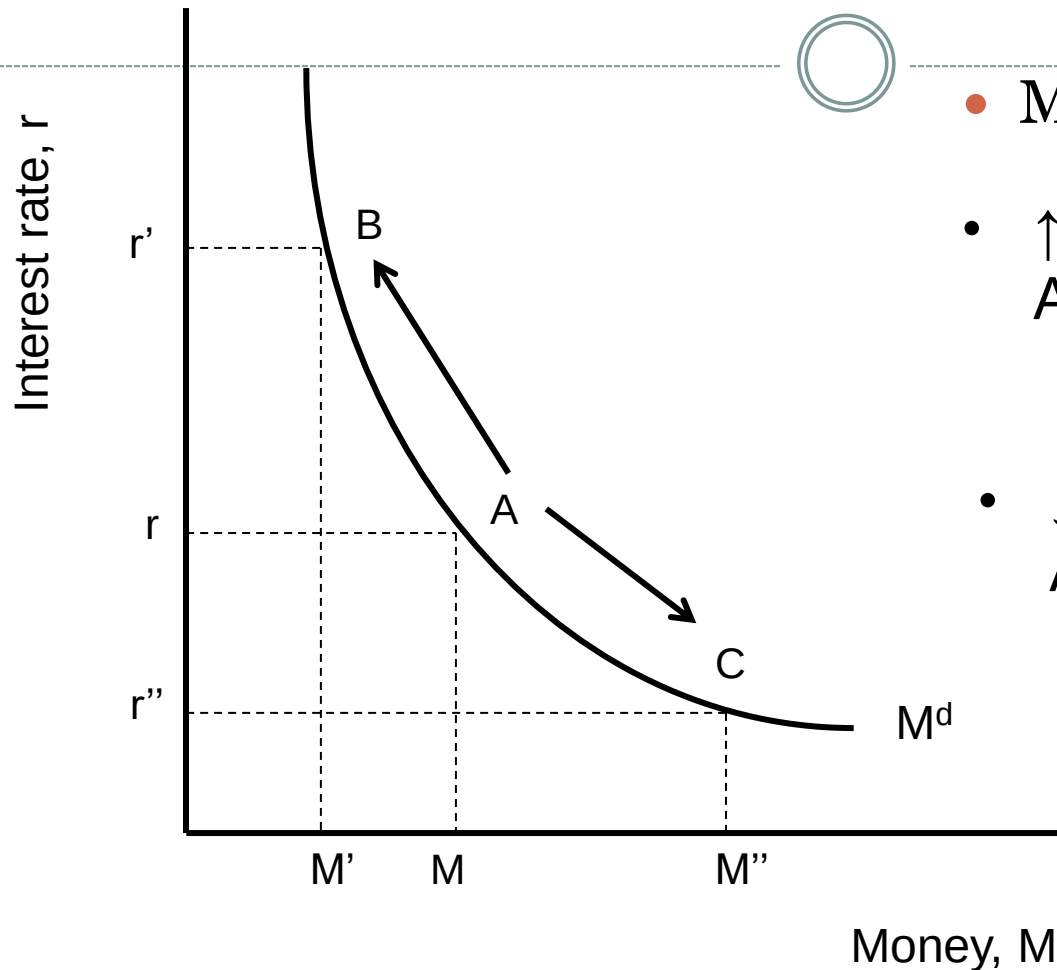
- Overall level of transaction in the economy (roughly proportional to nominal income)
- Interest rate

$$M^d = m(y, r)$$

(-)

The demand for money, M^d , is equal to nominal income, Y , times a function of the interest rate, i , with the function denoted by m

DERIVING THE DEMAND FOR MONEY (2)

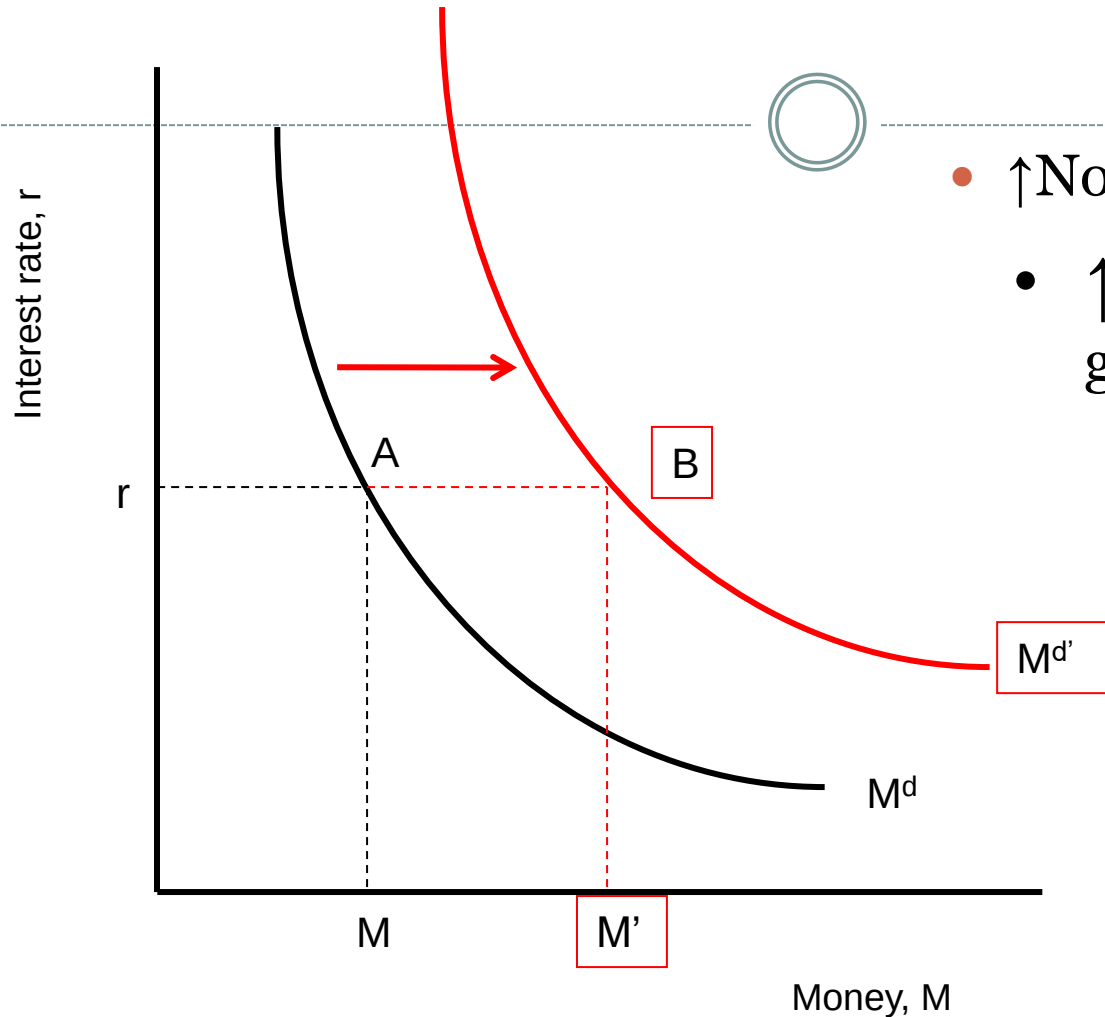


• $M^d \leftrightarrow -ve \leftrightarrow r$

• $\uparrow r \rightarrow \downarrow M^d$ (movement $A \rightarrow B$)

• $\downarrow r \rightarrow \uparrow M^d$ (movement $A \rightarrow C$)

DERIVING THE DEMAND FOR MONEY (3)

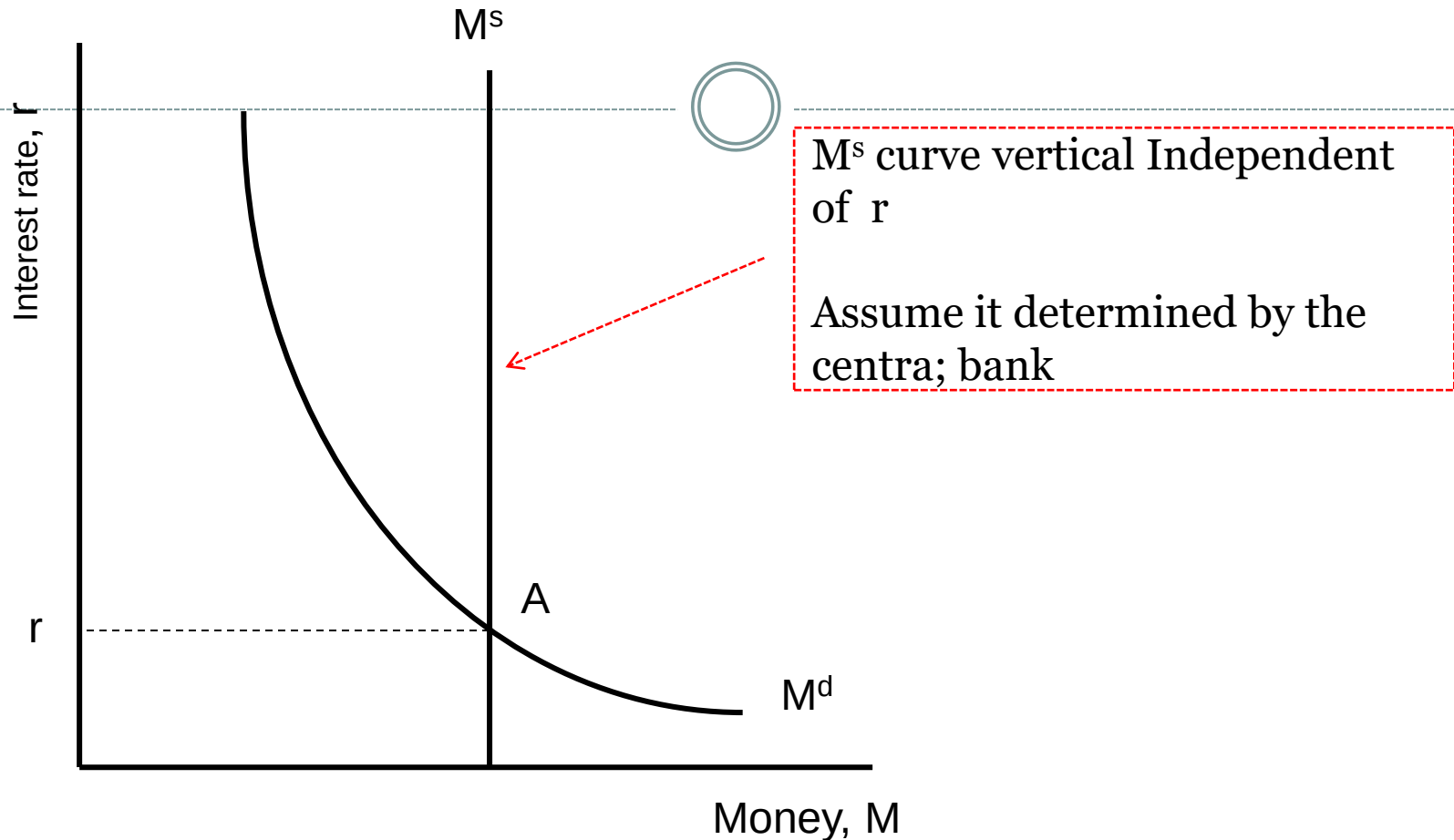


- $\uparrow$ Nominal Y (Y)?
- $\uparrow$ Nominal Y for a given i :
 - $\uparrow M^d$ (shift)

DETERMINATION OF THE INTEREST RATE I MONEY DEMAND, MONEY SUPPLY, AND THE EQUILIBRIUM INTEREST RATE

- $M^s = M$
- Equilibrium in the financial market is where $M^s = M^d$.
Using $M^s = M$ and $M^d = m(y, r)$, the equilibrium condition is:
- Money supply = money demand
- $M^s = (y, r)$
- Thus, the interest rate, r , must be such, that given income, Y , people are willing to hold an amount of money equal to the existing money supply, M^s
- This equilibrium relation is called the LM relation

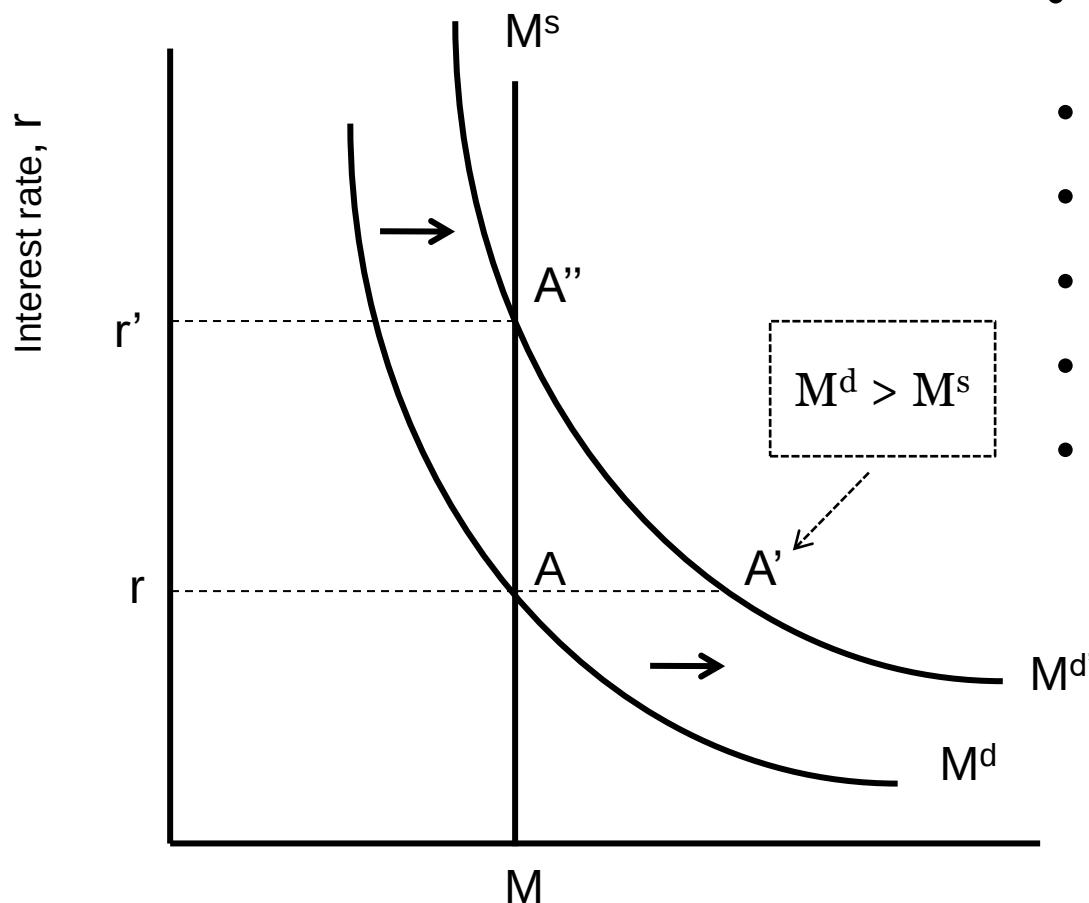
EQUILIBRIUM IN THE FINANCIAL MARKET (1)



- The interest rate must be such that the supply of money is equal to the demand for money

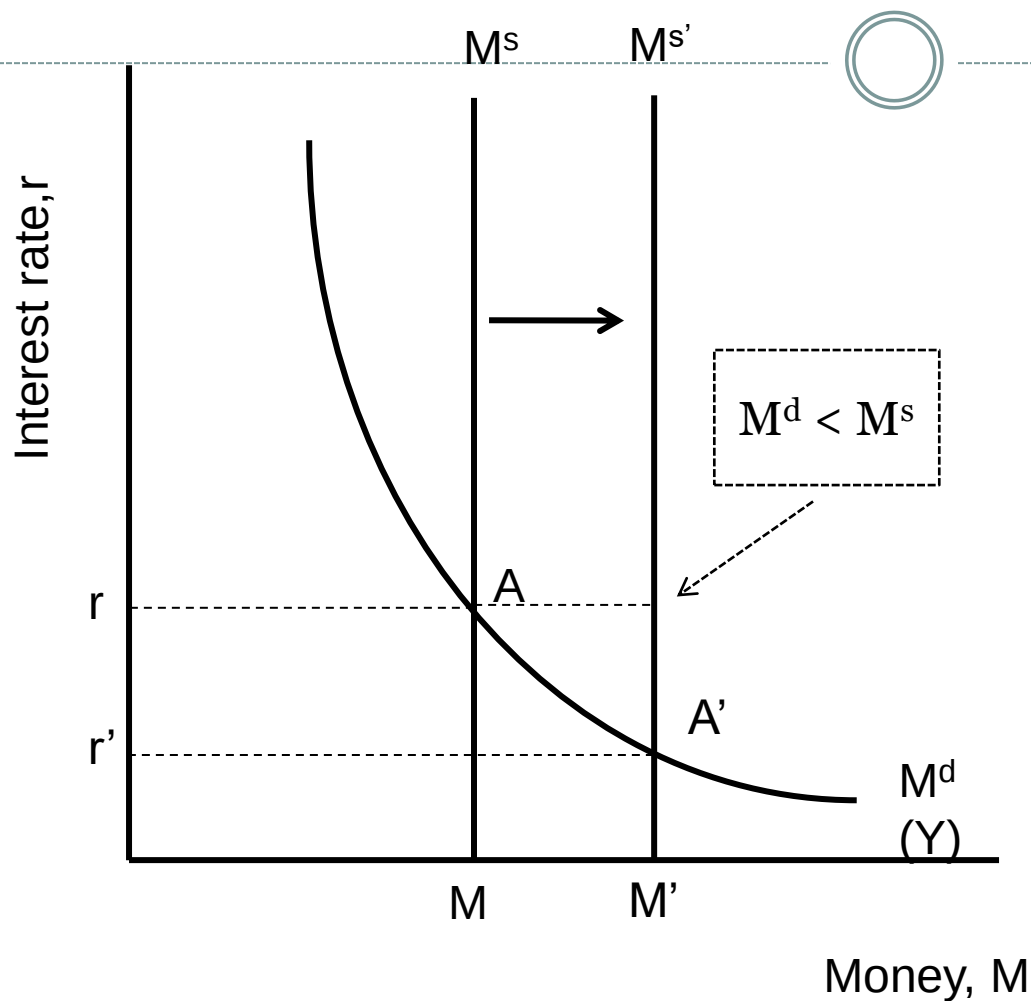
EQUILIBRIUM IN THE FINANCIAL MARKET (2)

- How do changes in nominal income or changes in money supply affect the equilibrium interest rate?



- $\uparrow$ Nominal Y
- $\uparrow$ level of transactions
- $\uparrow M^d$ at any r
- M^d curve shifts right
- Equilibrium A to A'
- $\uparrow$ equilibrium interest rate

EQUILIBRIUM IN THE FINANCIAL MARKET (3)

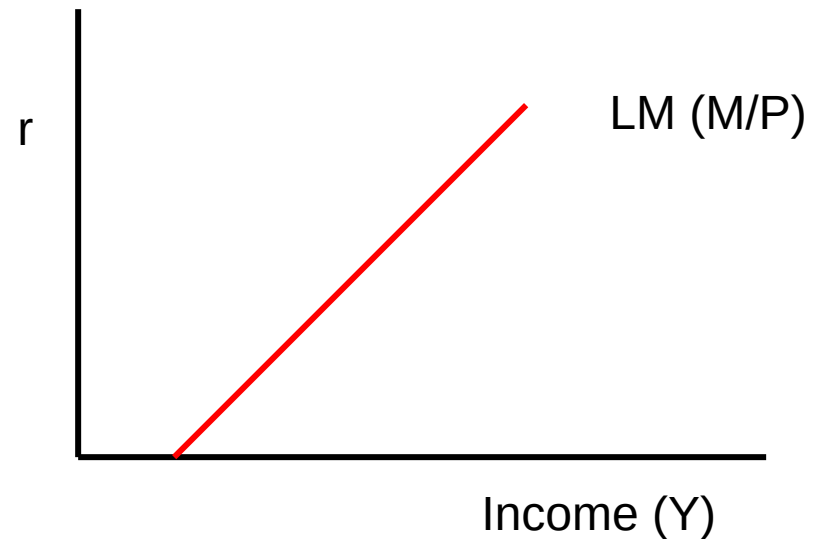


• $\uparrow M^s$

- Shift of M^s curve to the right
- Equilibrium from A to A'
- $\downarrow r$
- $\uparrow M^s \rightarrow \downarrow r$

LM CURVE

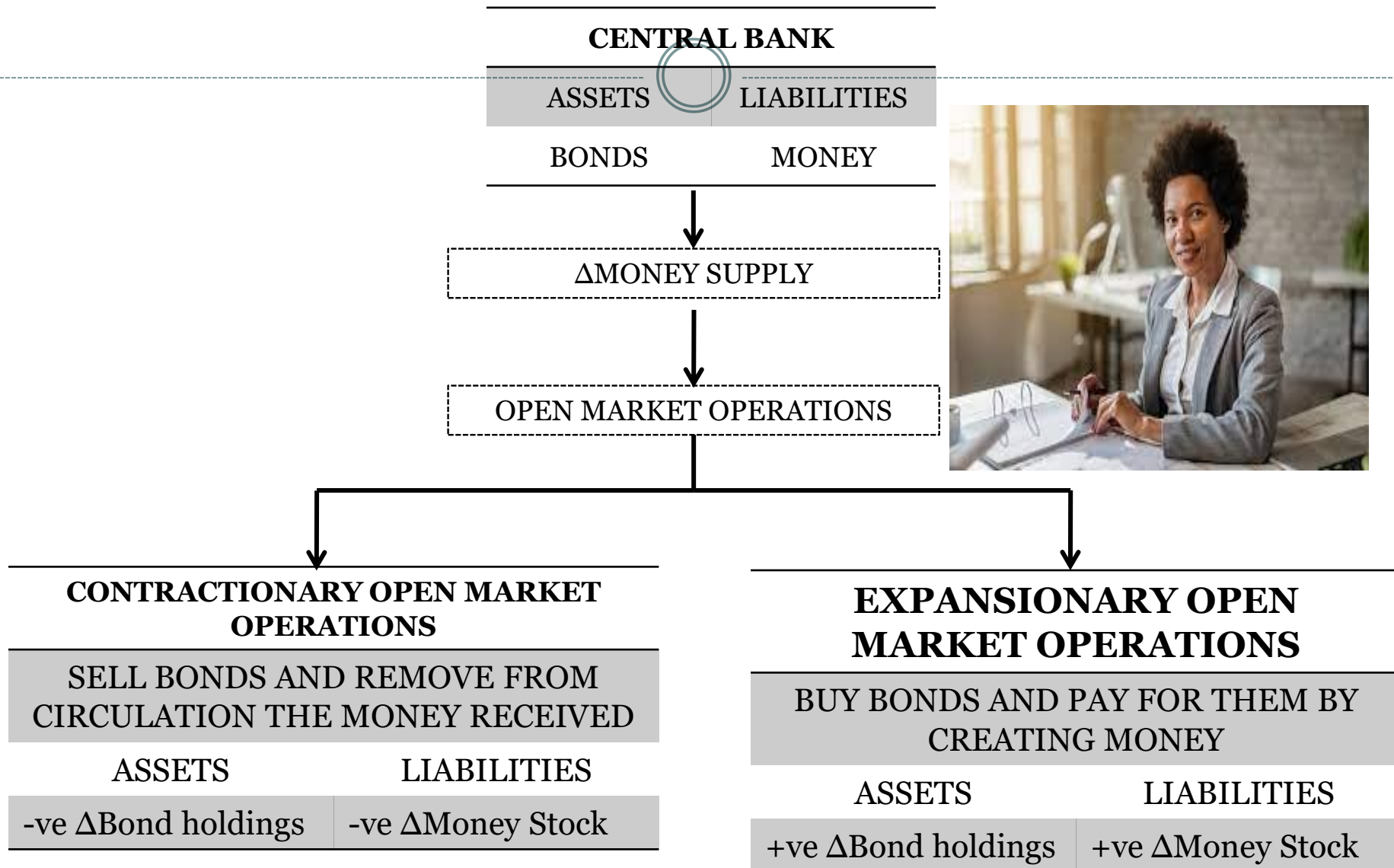
- The equilibrium in the money market is the basis for the LM
- The LM – shows various combinations of interest rate and income in which the money market is in equilibrium
- It's a function of real money supply (changes in real money supply will shift the LM curve inwards or outwards)



The slope of the LM curve represents the money demand responsiveness of M^d to the rate of interest. The smaller the responsiveness, the steeper the slope of the LM curve will be.

The greater the responsiveness of M^d to the interest rate, the flatter the LM curve will be.

OPEN MARKET OPERATIONS



BOND PRICES AND BOND YIELDS (1)

- **Objective:** you need to understand the **inverse relation** between the **price of a bond, P_B** , and the **interest rate, r** .

Rate of return on the one-year bond (i.e. Interest rate) $\rightarrow r = \frac{K100 - P_B}{P_B}$

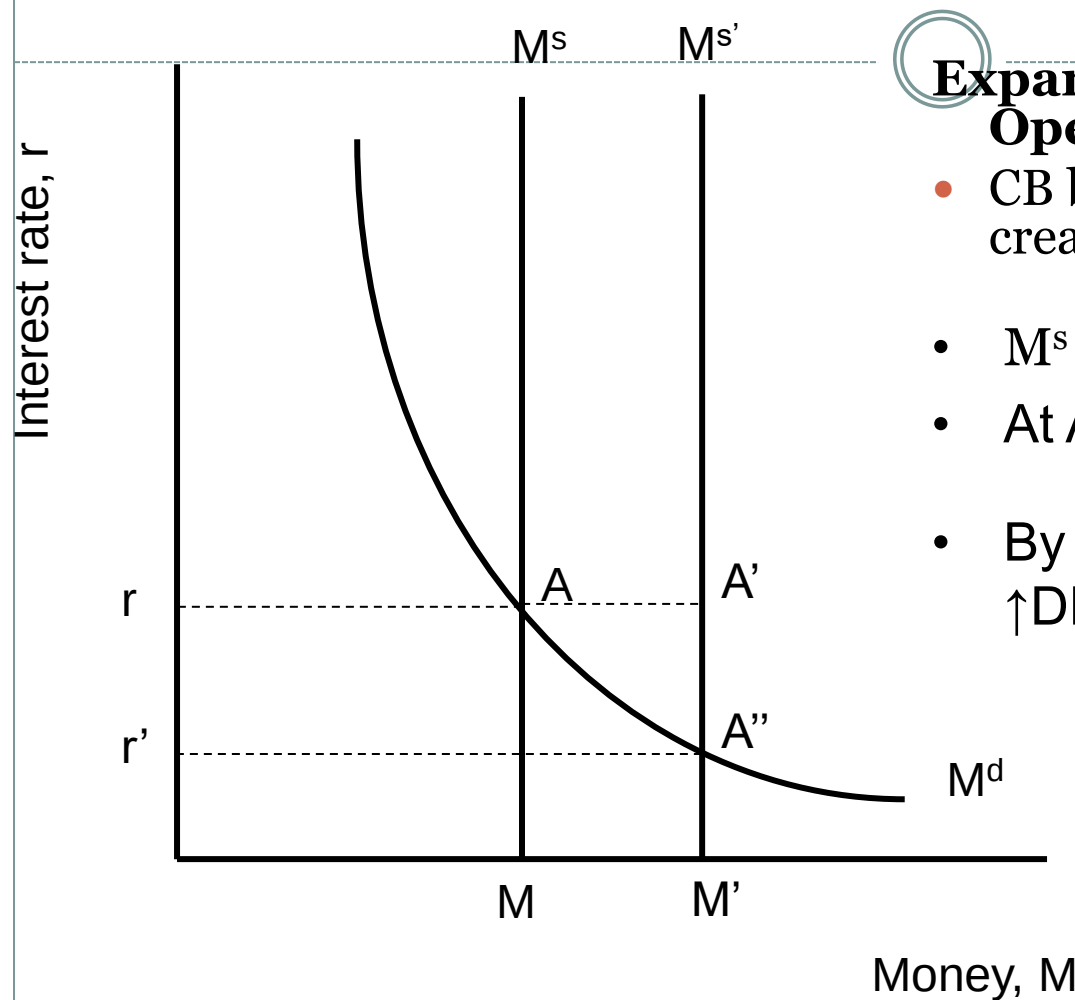
Future payment on one-year bond $\rightarrow K100$

Price of the bond today $\rightarrow P_B$

P_B	Calculation	r
K95	$(100-95)/95$	5.3%
K90	$(100-90)/90$	11.1%
k80	$(100-80)/80$	25%

- If the interest rate is positive then the price of the bond is less than the final payment
- Open market operations?

BOND PRICES AND BOND YIELDS (2)



Expansionary Open Market Operations

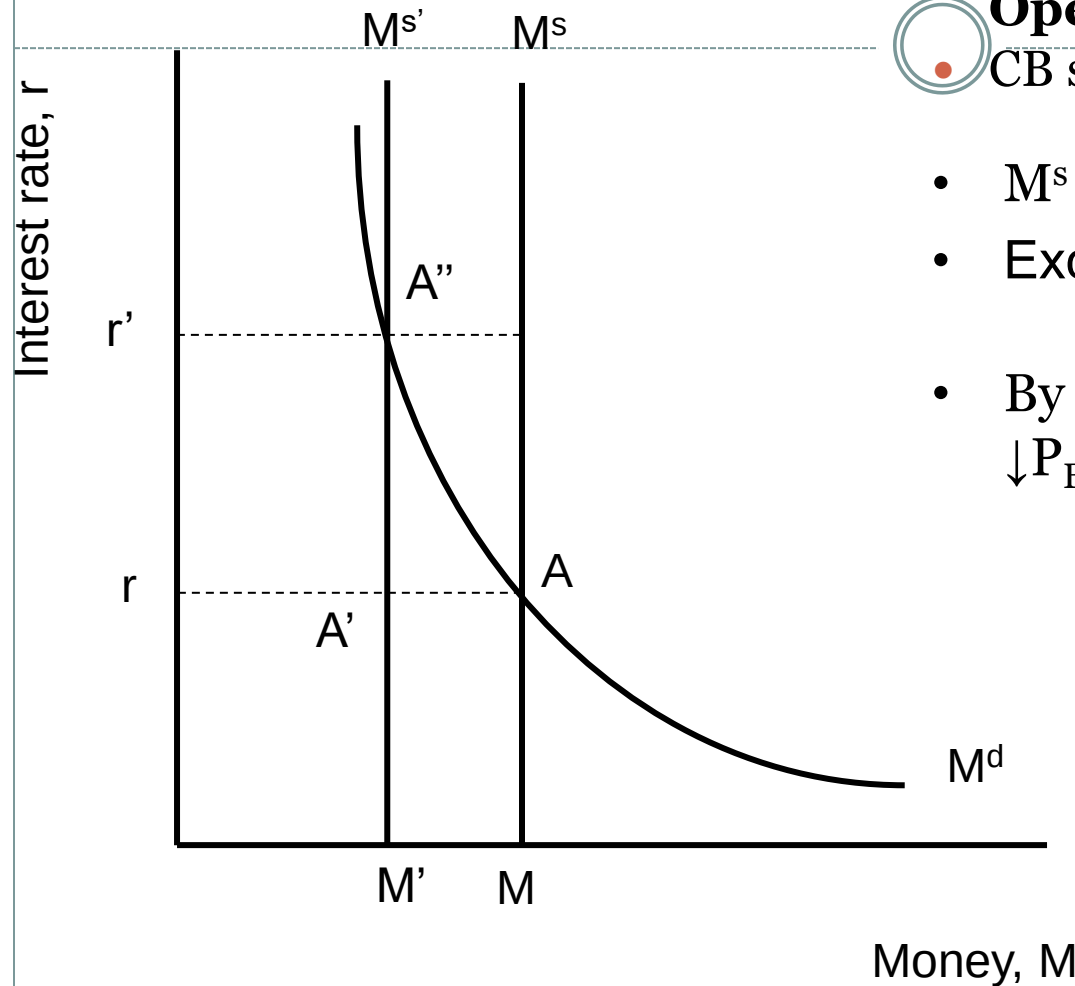
- CB buys bonds - pays for them by creating money
- M^s shifts right
- At A' excess supply money
- By buying bonds (hence $\uparrow M^s$) $\rightarrow$ $\uparrow DD_B \rightarrow \uparrow P_B \rightarrow \downarrow i$ ($M^s = M^d = \text{equil}^m$)

BOND PRICES AND BOND YIELDS (3)

Contractionary Open Market Operations:

• CB sells bonds in open market (i.e. $\downarrow M^s$)

- M^s curve shifts to the left
- Excess demand for money at A'
- By selling bonds (i.e. $\downarrow M^s$) $\rightarrow \uparrow SS_B \rightarrow \downarrow P_B \rightarrow \downarrow r$ ($M^s = M^d = \text{equil}^m$)



- Choosing **money** or choosing the **interest rate**?

THE DETERMINATION OF INTEREST RATE, II* WHAT BANKS DO



Extend model to include financial intermediaries, specifically, banks (banks liabilities = money).

A banks balance sheet is shown by:

ASSETS		LIABILITIES	
BANKS			
RESERVES		CHECKABLE DEPOSITS	
LOANS			
BONDS			
CENTRAL BANK			
BONDS		RESERVES	
		CURRENCY	

THE SUPPLY AND THE DEMAND FOR CENTRAL BANK MONEY

- Structure of the demand and the supply of central bank money

Demand for money

Demand for checkable deposits

Demand for currency

Demand for reserves by banks

Demand for central bank money

Supply of central bank money



THE DEMAND FOR MONEY

- When people can hold both currency and checkable deposits – the demand for money involves two decisions:
 1. How much money to hold (as opposed to bonds)?
 2. How much of this money to hold in currency (CU^d) and how much to hold in checkable deposits (D^d)?

$$M^d = m(y, r)$$

$$\text{Demand _ for _ currency : } CU^d = cM^d$$

$$\text{Demand _ for _ checkable _ deposits : } D^d = (1 - c)M^d$$

Modelling Money supply



- We have exogenous variable in the money supply process
- Monetary base: $B = C + R$
 - i.e., money in circulation plus reserves which are controlled by the central bank
- The central bank also regulates reserve ratio given as
- Reserve-deposit ratio, $rr = R/D$
- Household/individuals also determine how much to deposit based on their preferences
- Currency-deposit ratio $cr = c/D$

Money supply Process

- We can now solve for the money supply in terms of the exogenous variables describe
- We know that:
- $M = C + D$ Multiply by 1 (B/B)
- $M = C + D = \frac{C + D}{B} \times B = m \times B$
- Where $m = \frac{C + D}{B}$

Money supply equation



- Then we get

$$m = \frac{C + D}{C + R} = \frac{(C/D) + (D/D)}{(C/D) + (R/D)} = \frac{cr + 1}{cr + rr}$$

- The Money multiplier

- Recall $M = m + B$ where

- If $rr < 1$, then $m > 1$

$$m = \frac{cr + 1}{cr + rr}$$

- If monetary base changes by ΔB

- The $\Delta M = m + \Delta B$

Example



- Suppose the monetary base is K800, the reserve deposit ratio is 0.1 and the currency deposit ratio is 0.8
- What is the money supply?
- We know that $M = mB$ and the money multiplier is:

$$m = \frac{cr + 1}{cr + rr}$$

- The $m = \frac{0.8+1}{0.8+0.1} = 2$
- The $M = 2.0 \times 800 = ZMW1,600$

Overall determinants of money supply



- a. Central bank
- b. Banks (depository institutions; financial intermediaries)
- c. Depositors (individuals/HH and institutions)
- d. Borrowers (individuals and institutions)

Question



Suppose depositors decide to hold more of their money as currency and less in the form of demand deposits.

1. Determine impact on money supply.
 2. Explain the intuition for your result.
- ❑ Impact of an increase in the currency-deposit ratio $\Delta cr > 0$.
1. An increase in cr increases the denominator of m proportionally more than the numerator. So the multiplier - m falls, causing M – money supply to fall too.
 2. If depositors deposit less of their money then banks the credit creation process is slowed down that is banks can't make as many loans, so the banking system won't be able to “create” as much money.

The instruments of monetary policy



- **Monetary policy** process by which the monetary authority of a country controls the supply and availability of money
 - often targeting a rate of interest for the purpose of promoting economic growth and stability
- There are three instruments of monetary policy
 - a) Open market operations
 - b) Reserve requirements
 - c) The discount rate

TOOLS OF MONETARY POLICY



1. Open-Market Operations

- “open market operations” are the buying and selling of government bonds by the central bank
- Operations are “open” in the sense that anyone---can buy or sell in this market.
- how it works:
 - If the central bank buys bonds or securities from the public, it pays with new currency, increasing B and therefore M
 - Banks will have increased reserves

TOOLS OF MONETARY POLICY



2. Reserve requirement

- Reserve Requirements influence the ability of banks to create new loans which affects the broader aggregates (M_1, M_2, M_3)
- where central bank regulations that require banks to hold a minimum reserve-deposit ratio
- how it works: the reserve requirements affect **rr** and **m**
 - If the reduces reserve requirements, then banks can make more loans and “create” more money from each deposit.
 - Money supply increasing
- If it raises the Reserve Ratio
 - Banks must hold more reserves
 - Banks decrease lending
 - Money supply decreases

TOOLS OF MONETARY POLICY



3. The discount rate

- The interest rate that the CB charges on loans it makes to banks.
- *how it works:*
 - When banks borrow from the CB, their reserves increase, allowing them to make more loans and “create” more money
 - The CB can increase **B** by lowering the discount rate to induce banks to borrow more reserves from the CB.

Frequency of instrument use

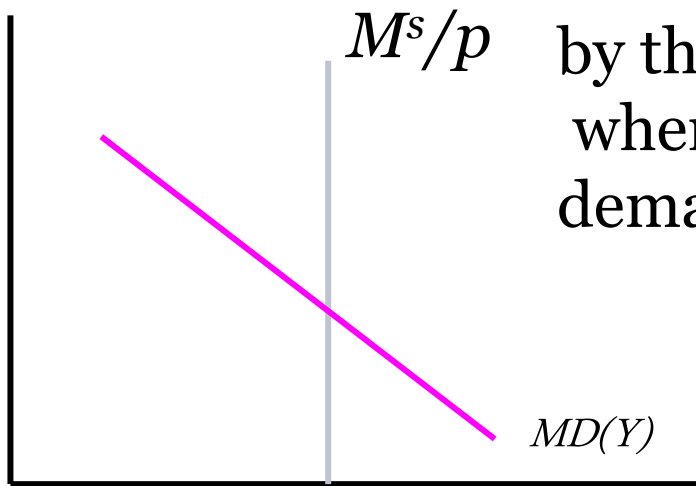


- a. Open market operations:** most frequently used .
- b. Changes in reserve requirements:** least frequently used.
 - ✦ Making them too low creates a risk of bank runs.
Making them too high makes banking unprofitable.
 - ✦ In addition, banking would be difficult if the CB changed reserve requirements frequently.
- c. Changes in the discount rate:** This one is largely symbolic; the CB is a “lender of last resort,” hence it does not usually make loans to banks on demand.

FISCAL AND MONETARY POLICIES

The demand for money = $m = m(y, r)$: money demand function

$$m_y > 0 \text{ and } m_r < 0$$



We assume money supply is determined by the central bank. Equilibrium obtains when money market when money demand